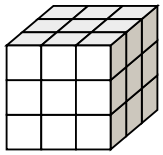
Preview: y2007_q17 (v1)

Painted cube cut up: corners 3 faces, edges 2, face middles 1, inside 0

Part A: Each question is worth 3 points

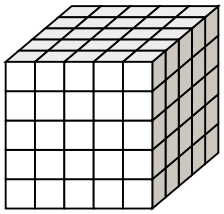
1. A cube with sides of 3 cm is painted grey all over and then cut into small cubes with sides of 1 cm.



How many of the small cubes have exactly 2 painted faces?

A. 36 ***B.*** 12 ***C.*** 8 ***D.*** 6 ***E.*** 4

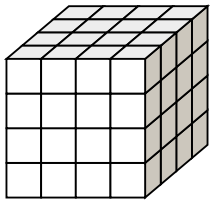
2. A cube with sides of 5 cm is painted grey all over and then cut into small cubes with sides of 1 cm.



How many of the small cubes have exactly 2 painted faces?

A. 12 ***B.*** 18 ***C.*** 24 ***D.*** 36 ***E.*** 60

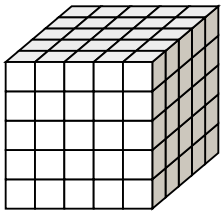
3. A big cube made of $4 \times 4 \times 4 = 64$ small cubes is painted on the outside, also underneath, and then taken apart.



How many small cubes have exactly 3 painted faces?

A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 12 ***E.*** 16

4. A big cube made of $5 \times 5 \times 5 = 125$ small cubes is painted on the outside, also underneath, and then taken apart.

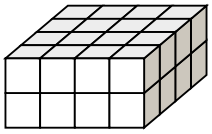


How many small cubes have exactly 3 painted faces?

A. 4 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

Part B: Each question is worth 4 points

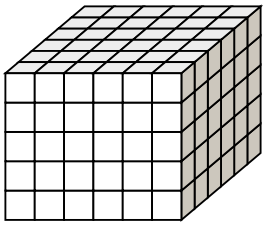
5. A brick 4 cubes long, 4 cubes deep and 2 cubes tall is made of small cubes. It is painted on the outside, also underneath, and then taken apart.



How many small cubes have exactly 2 painted faces?

A. 20 ***B.*** 19 ***C.*** 16 ***D.*** 8 ***E.*** 0

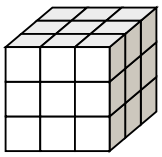
6. A brick 6 cubes long, 6 cubes deep and 5 cubes tall is made of small cubes. It is painted on the outside, also underneath, and then taken apart.



How many small cubes have exactly 1 painted face?

A. 98 ***B.*** 84 ***C.*** 80 ***D.*** 48 ***E.*** 44

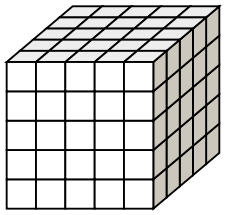
7. A cube made of $3 \times 3 \times 3$ small cubes stands on the table and is painted grey on all sides you can reach: the top and the four sides, but **not** the bottom. Then it is taken apart.



How many small cubes have exactly 2 painted faces?

A. 8 ***B.*** 10 ***C.*** 12 ***D.*** 14 ***E.*** 16

8. A cube made of 5 × 5 × 5 small cubes stands on the table and is painted grey on all sides you can reach: the top and the four sides, but **not** the bottom. Then it is taken apart.



How many small cubes have exactly 3 painted faces?

A. 8 ***B.*** 6 ***C.*** 5 ***D.*** 4 ***E.*** 2

Part C: Each question is worth 5 points

9. A big cube made of small cubes is painted on the outside and taken apart. Exactly 36 small cubes have 2 painted faces.
How many small cubes was the big cube made of?

A. 27 ***B.*** 64 ***C.*** 72 ***D.*** 100 ***E.*** 125

10. A big cube made of small cubes is painted on the outside and taken apart. Exactly 24 small cubes have 2 painted faces.
How many small cubes was the big cube made of?

A. 96 ***B.*** 64 ***C.*** 48 ***D.*** 27 ***E.*** 8

Answer Key

1. **B**	2. **D**	3. **C**	4. **E**	5. **C**	6. **C**
7. **C**	8. **D**	9. **E**	10. **B**		

Solutions

1. B (12) — They sit on the edges, but not at the corners. 12 edges with 3 – 2 = 1 such cubes each: 12 × 1 = 12. [2007 Q17 · two_faces]

2. D (36) — They sit on the edges, but not at the corners. 12 edges with 5 – 2 = 3 such cubes each: 12 × 3 = 36. [2007 Q17 · two_faces]

3. C (8) — Only the corner cubes have 3 painted faces, and a cube always has 8 corners, however big it is. [2007 Q17 · always_eight]

4. E (8) — Only the corner cubes have 3 painted faces, and a cube always has 8 corners, however big it is. [2007 Q17 · always_eight]

5. C (16) — On each edge, the cubes between the two corners: the brick has 4 edges of each length 4, 4, 2, so 4 × (2 + 2 + 0) = 16. [2007 Q17 · painted_brick]

6. C (80) — The middle part of each face: 2 × (4 × 4 + 4 × 3 + 4 × 3) = 80. [2007 Q17 · painted_brick]

7. C (12) — Top edges without their corners: 4 × 1 = 4 (top and one side). Upright edges without the top corner: 4 × 2 = 8 (two sides; the bottom corners count here because their bottom is not painted). Total 12. [2007 Q17 · not_underneath]

8. D (4) — Only the 4 top corners: they have the top and two sides painted. The bottom corners lose their bottom face. [2007 Q17 · not_underneath]

9. E (125) — A cube has 12 edges, so each edge has 36 ÷ 12 = 3 cubes between its corners. With the 2 corners, an edge is 5 cubes long, and the cube has 5 × 5 × 5 = 125. [2007 Q17 · how_big]

10. B (64) — A cube has 12 edges, so each edge has 24 ÷ 12 = 2 cubes between its corners. With the 2 corners, an edge is 4 cubes long, and the cube has 4 × 4 × 4 = 64. [2007 Q17 · how_big]